Optimised Tree Search vs Lowest Priority First Heuristic (LPFH): Move Reduction by Bay Setting

Oct 15, 2026 · @Dustin

Preliminary Result

Optimised tree search needs 13–19% fewer weekend moves than LPFH when every pickup date is known, and 7–10% fewer when half the dates are unknown. LPFH is assumed here to be the operator's current practice.

Bay setting	Bays compared	Weekend moves, LPFH (avg)	Weekend moves, Optimised (avg)	Fewer weekend moves	95% range of per-bay saving	Optimised better / same / worse
20x8, all dates known, exact	20 of 20	79.5	64.7	**18.6%**	14.0% to 21.4%	19 / 0 / 1
20x8, all dates known, 3-day windows	20 of 20	76.1	64.7	**15.0%**	11.8% to 17.6%	19 / 0 / 1
20x8, half known, exact	20 of 20	43.8	40.0	**8.5%**	6.2% to 9.4%	18 / 2 / 0
20x8, half known, 3-day windows	20 of 20	41.9	39.0	**6.9%**	4.7% to 7.8%	17 / 3 / 0
5x8, all dates known, exact	12 of 20	23.2	20.2	**12.9%**	5.9% to 16.1%	10 / 1 / 1
5x8, all dates known, 3-day windows	13 of 20	24.2	20.2	**16.8%**	7.8% to 21.5%	11 / 0 / 2
5x8, half known, exact	10 of 20	11.2	10.1	**9.8%**	1.5% to 11.5%	4 / 6 / 0
5x8, half known, 3-day windows	10 of 20	10.1	9.4	**6.9%**	−2.3% to 10.4%	2 / 6 / 2

"Bays compared" counts the bays where LPFH found a plan; Optimised tree search found one on every bay.

How it was measured, and caveats

Both methods planned the weekend reshuffle on the same bays, and the table compares the number of moves in each plan.

- **Paired bays:** 20 synthetic bays per setting (seeds 0–19), 75% full. Both methods ran on each bay with a 60-second limit.
- **Fewer weekend moves:** (LPFH average − Optimised average) ÷ LPFH average. The 95% range is a bootstrap interval for the average per-bay saving.
- **LPFH failures:** on the 5x8 bays LPFH found no plan on 35 of 80 bays, so those rows compare only the bays it solved. These may be the easier ones.
- **Assumed baseline:** LPFH (lowest-priority-first, Expósito-Izquierdo et al. 2012) stands in for the operator's practice. Cogent's actual practice is not yet known.
- **Synthetic data:** the bays and dates are generated, not taken from Cogent's yard. Treat all figures as illustrative until calibrated.
- **Weekend only:** the table counts weekend moves, not the weekday relocations that follow each plan.